

G H RAISONI COLLEGE OF ENGINEERING AND MANAGEMENT

(An Empowered Autonomous Institute affiliated to SPPU)

(NAAC Accredited With A+ Grade)

WAGHOLI, PUNE – 412207

Department of CSE (Data Science)

Cleaning and Analyzing Raw Daily Mandi Commodity Price Data to Improve Agricultural Market Price Intelligence and Forecasting

TAE-2

Subject: Business Intelligence

Guided By:

Mr. Chinmay Mukim

Team Members

Devyani Daphal — (P-13)

Devyani Shinde — (D-65)

Project Overview

- Analyzes commodity market prices across State, District, Market, Commodity, Variety and Grade
- Dataset includes Minimum, Maximum and Modal price, arrival date, location and commodity information
- Instead of directly charting raw data, a complete pipeline was built: cleaning → transformation → warehouse → SQL → visualization



Problem Statement

- Prices vary across markets, states, commodities, varieties and grades
- Raw data is hard to analyze directly because of inconsistent column names, text formatting issues, mixed data types and large record volumes
- To develop a data analytics system that cleans, organizes, stores, analyzes and visualizes commodity market price data

Objectives

- Clean and preprocess raw commodity market data
- Perform ETL using Python
- Create a data warehouse using MySQL
- Implement a star schema using fact and dimension tables
- Perform analytical queries using SQL
- Calculate useful metrics such as price spread
- Create an interactive Power BI dashboard
- Generate business-oriented insights
- Provide a foundation for future price forecasting using historical multi-date data

Dataset & Data Understanding

Column	Meaning
State / District / Market	Location of the market (mandi)
Commodity / Variety / Grade	What is being traded and its quality
Arrival Date	Date of market arrival
Min / Max / Modal Price	Minimum, maximum and most representative traded price

Derived Columns

- Price Spread = Max Price – Min Price
- Price Spread % = $((\text{Max} - \text{Min}) / \text{Modal}) \times 100$
- Year, Month, Day extracted from arrival date

Why modal price?

Modal price represents the price occurring most commonly in the market data and is useful as a representative market price.

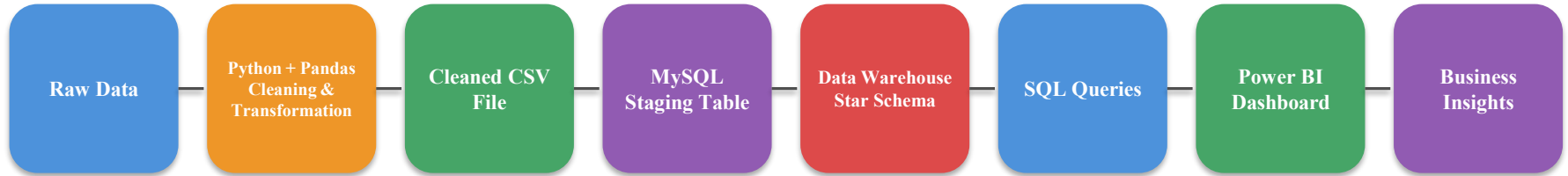
Challenges Faced

- Raw data formatting — column names contained artifacts like `_x0020_` and were standardized
- Inconsistent text — unwanted leading/trailing spaces were removed using Python string operations
- Data type conversion — dates converted to datetime; prices converted to numeric
- Warehouse design — flat table converted into a star schema (1 fact + 5 dimensions)
- MySQL ↔ Power BI — configured MySQL connector so the warehouse could power the dashboard
- Forecasting limitation — data has only one arrival date (28 Jul 2026), not enough for time-series forecasting

Technical Stack

Technology	Purpose
Python	Data preprocessing and ETL
Pandas	Data loading, cleaning and transformation
NumPy	Numerical operations
Jupyter Notebook	ETL development and experimentation
MySQL	Data warehouse
MySQL Workbench	Database creation, SQL and data loading
SQL	Data analysis and aggregation
Power BI	Dashboard and visualization
GitHub	Project / version sharing

End-to-End Implementation Workflow



The project follows an ETL-to-dashboard architecture: raw data is extracted and cleaned using Python, loaded into a staging table, transformed into dimensional warehouse tables in MySQL, analyzed using SQL, and finally connected to Power BI for visualization.

ETL / Data Preprocessing

Extract

- Loaded raw CSV using Pandas: `pd.read_csv(...)`
- Profiled rows, columns, dtypes, missing values, duplicates

Transform

- Standardized column names (removed `_x0020_` artifacts)
- Stripped whitespace from text columns
- Converted `arrival_date` to datetime
- Converted min/max/modal price to numeric
- Created `price_spread` and `price_spread_pct`
- Extracted year, month, day

Load

- Saved as `commodity_cleaned.csv`
- Loaded into the MySQL staging table

Data Warehouse & Star Schema

Why star schema?

A star schema separates measurable business data (facts) from descriptive information (dimensions). This makes analytical queries easier to understand and supports efficient reporting.

Easiest way to remember it:

**Dimension = WHO / WHAT /
WHERE / WHEN**

Fact = HOW MUCH



Fact & Dimension Tables

fact_market_price

Keys:

price_id, date_key, location_key,
 commodity_key, variety_key,
 grade_key

Measures:

min_price, max_price, modal_price,
 price_spread, price_spread_pct

Why is this a fact table?

It contains measurable numerical business values
 (prices, spread) plus foreign keys connecting to
 descriptive dimensions.

Dimension	Contains
dim_date	date_key, full_date, year, month, day
dim_location	location_key, state, district, market
dim_commodity	commodity_key, commodity
dim_variety	variety_key, variety
dim_grade	grade_key, grade

SQL Analysis & Business Questions

Business Question	Visualization
Which commodities have higher average modal prices?	Bar chart
Which markets have higher average modal prices?	Bar chart
Which commodities have larger price spreads?	Bar chart
How does average price vary by state?	Bar chart / map
How do min, max and modal prices compare?	Clustered column chart
How do varieties differ in average modal price?	Bar chart
Does grade relate to price and price spread?	Grouped summary (count, avg price, avg spread)

Sample SQL: `SELECT c.commodity, ROUND(AVG(f.modal_price),2) AS avg_modal_price FROM fact_market_price f JOIN dim_commodity c ON f.commodity_key = c.commodity_key GROUP BY c.commodity ORDER BY avg_modal_price DESC;`

Power BI Dashboard / Visualization

Purpose

Power BI converts the structured warehouse data into an interactive dashboard, connected directly to the MySQL data warehouse.

KPI Cards

Average Modal Price

Average Minimum Price

Average Maximum
Price

Average Price Spread

Charts

Avg Modal Price by Commodity (Bar)

Easy comparison between categories

Avg Modal Price by Market (Bar/Column)

Compares market-level prices

Price Spread by Commodity (Bar)

Shows min–max price gap by commodity

State-wise Avg Price (Bar / Map)

Shows geographical variation

Min vs Modal vs Max (Clustered Column)

Direct comparison of 3 price measures

Business Insights

Insert your actual SQL / Power BI results into this structure:

Commodity

[Commodity A] has the highest average modal price analyzed, while [Commodity B] has a lower average.

Market

[Market A] has the highest average modal price among the analyzed markets.

Price Spread

[Commodity A] has the highest average price spread, indicating a larger min–max price gap.

State

State-level analysis shows variation in average modal prices across states.

Grade

The grade analysis compares average modal price and price spread across grades.

Results, Limitations & Future Scope

Results Achieved

- Raw data cleaned & quality checked
- Derived price metrics created
- ETL pipeline implemented
- MySQL star-schema warehouse built
- SQL analytical queries answered
- Power BI dashboard developed

Limitations

- Dataset has only one arrival date — no reliable trend analysis
- Meaningful forecasting needs multiple historical observations
- Current analysis is descriptive and comparative, not predictive
- External factors such as weather, demand and seasonality are not included

Future Scope

- Multi-year/multi-month data for time-series forecasting
- ARIMA / SARIMA / Prophet / ML forecasting models
- Integrate weather, demand-supply and transport cost data
- Historical Data → Feature Engineering → Forecasting Model → Power BI